Binomial Theorem

#Factorial Notation :

$$n! = \lfloor n = 1 \times 2 \times 3 \times 4 \cdots n$$

($n \in$ ~~Whole~~ Whole Numbers)

→ $n! = (n-1)! \times n \Rightarrow (n-1)! = \frac{n!}{n}$

→ $0! = 1$ always (Also, $1! = 1$)

#Binomial Coefficients :

(Other Representations)

$$^nC_r = \frac{n!}{r!\,(n-r)!} = \binom{n}{r} = C_r$$

- (^nC_r) is always a +ve quantity
- $n, r \in$ Whole Numbers.
- For ^nC_r to be defined, $n \geq r$

Important Values ⇒

① $^nC_0 = 1$ ② $^nC_1 = n$ ③ $^nC_2 = \frac{n(n-1)}{2}$

Properties of Binomial Coefficient :→

1) $^nC_r = {}^nC_{n-r}$

2) $^nC_r + {}^nC_{r-1} = {}^{n+1}C_r$ (Upar aur neeche, dono gagan baaf valf term)

3) $\frac{^nC_r}{^nC_{r-1}} = \frac{n-r+1}{r}$ (Bada wala 'r' as it is utha ke le Khna hai)

$\frac{^nC_{r+1}}{^nC_r} = \frac{n-(r+1)+1}{r+1} \Rightarrow \frac{n-r}{r+1}$

$\frac{^nC_{r+2}}{^nC_{r+1}} = \frac{n-(r+2)+1}{r+2} \Rightarrow \frac{n-r-1}{r+2}$

4) $^nC_r = \frac{n}{r}\,(^{n-1}C_{r-1})$

#Binomial Expression :

→ Any expression consisting of only two terms.

1) $(x^{100}+y)$ ✓ 2) $x+4x+y \Rightarrow 5x+y$ ✓

3) $(x+y+z)\,x$ 4) $x+8x = 9x\,(x)$

Binomial Theorem For Natural Index :→

$$(a+x)^n = {}^nC_0\,a^n x^0 + {}^nC_1\,a^{n-1}x^1 + {}^nC_2\,a^{n-2}x^2 + {}^nC_3\cdot a^{n-3}\cdot x^3 + \cdots$$

$$\Rightarrow \boxed{(a+x)^n = \sum_{r=0}^{n} {}^nC_r\,a^{n-r}\,x^r}$$

(Position of a & x has significance)

→ Total Number of Terms in Binomial Expansion $= (n+1)$

→ Coefficient of x^r term in $(1+x)^n = {}^nC_r$

→ General term for $(a+x)^n$

$$T_{r+1} = {}^nC_r\,(a)^{n-r}\,(x)^r$$

→ General term for $(1+x)^n$.

$$T_{r+1} = {}^nC_r \cdot x^r$$

TYPE-1 : Problems Involving sigma ⇒

Ques: $\sum_{r=0}^{n} (-1)^r \cdot {}^nC_r \left[\left(\frac{1}{2}\right)^r + \left(\frac{3}{4}\right)^r + \left(\frac{7}{8}\right)^r + \text{-- m terms}\right]$

Find the sum.

Soln $\left(\frac{1}{2}\right)^r + \left(\frac{3}{4}\right)^r + \cdots = \left(\frac{2^1-1}{2^1}\right)^r + \left(\frac{2^2-1}{2^2}\right)^r + \cdots$

$\times(-1)^r$ $= \left(\frac{-1}{2}\right)^r + \left(\frac{-3}{4}\right)^r + \cdots \left(-\frac{2^m-1}{2^m}\right)^r$

$\times\,{}^nC_r$ $= {}^nC_r\left(\frac{-1}{2}\right)^r + \cdots {}^nC_r\left(-\frac{2^m-1}{2^m}\right)^r$

We know,

$$\sum_{p=0}^{q} {}^qC_p\,(X)^p = (1+X)^q$$

So,

$$\sum_{r=0}^{n} {}^nC_r\left(\frac{-1}{2}\right)^r = \left(1-\frac{1}{2}\right)^n \Rightarrow \frac{1}{2^n}$$

$$\sum_{r=0}^{n} {}^nC_r\left(\frac{-3}{4}\right)^r = \left(1-\frac{3}{4}\right)^n = \frac{1}{(2^2)^n}$$

$$\sum_{r=0}^{n} {}^nC_r\left(\frac{-2^m-1}{2^m}\right)^r = \left(1-\frac{2^m-1}{2^m}\right)^n = \frac{1}{(2^m)^n}$$

Hence,

$$\text{Sum} = \underbrace{\frac{1}{2^n} + \frac{1}{(2^n)^2} + \frac{1}{(2^n)^3} + \cdots \frac{1}{(2^n)^m}}_{\text{GP where, } a = 1/2^n \quad r = 1/2^n}$$

Total terms $= m$

So, $\left[\text{Sum of GP} = \frac{a(1-r^n)}{(1-r)}\right]$

$$\Rightarrow \boxed{\frac{\frac{1}{2^n}\left(1-\frac{1}{2^{nm}}\right)}{\left(1-\frac{1}{2^n}\right)}} \quad \text{Ans}$$

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Ques $A = \sum_{K=0}^{n} (-1)^K \cdot {}^nC_K \left[\left(\frac{1}{2}\right)^K + \left(\frac{3}{4}\right)^K \cdots \left(\frac{31}{32}\right)^K\right]$

If $63A = 1 - \frac{1}{2^{\cancel{100}\,80}}$, then $n =$ ___.

Soln $A = \frac{\frac{1}{2^K}\left(1-\frac{1}{2^{5K}}\right)}{\left(1-\frac{1}{2^K}\right)}$ [Total terms = 5]

Now, $63A = 1 - \frac{1}{2^{80}}$

$\frac{63}{2^n}\left(1-\frac{1}{2^{5n}}\right) = \left(1-\frac{1}{2^n}\right)\left(1-\frac{1}{2^{30}}\right)$

On comparing, $5n = 30$

$\Rightarrow \boxed{n = 6}$ Ans

Type-2 : Problems Involving $(a+1)^{th}$ term ⇒

Ques 3rd term of expansion of $\left(1 + x^{\log_2 x}\right)^5$ is 2560. Find possible x.

Soln $T_{r+1} = 5C_r\,(1)^{5-r}\,(x^{\log_2 x})^r$

$\Rightarrow T_3 = 2560 = {}^5C_2\,(x^{\log_2 x})^2$

$2560 = \frac{5\times4^2}{2}\,(x^{\log_2 x})^2$

$\Rightarrow x^{\log_2 x} = \pm 16$

Taking $\log_2$ both sides

$\log_2\left(x^{\log_2 x}\right) = \log_2(16)$ (-ve values not allowed in log)

$(\log_2 x)(\log_2 x) = 4$

$\Rightarrow \log_2 x = \pm 2$

$x = 4$ $\qquad$ $x = 1/4$ Ans

Ques: 4th term in expansion of $\left(\sqrt{\frac{1}{x^{1+\log x}}} + \frac{1}{x^{12}}\right)^6$ is 200 & $x > 1$, Find x.

Soln $T_{r+1} = {}^6C_r\left(\sqrt{x^{\frac{1}{1+\log x}}}\right)^{6-r} \cdot \left(\frac{1}{x^{12}}\right)^r$

$T_{3+1} = 200 = {}^6C_3\left(\sqrt{x^{\frac{1}{1+\log x}}}\right)^{6-3}\left(\frac{1}{x^{12}}\right)^3$

$200 = \frac{6\times5\times4}{1\times2\times3}\,(x)^{\frac{3}{2(1+\log_{10}x)}} \cdot (x)^{1/4}$

$\Rightarrow (x)^{\frac{3}{2(1+\log x)} + \frac{1}{4}} = 100$

$\left(\frac{3}{2(1+\log x)} + \frac{1}{4}\right)(\log x) = \log(10)$

Put $\log x = t$

$\left(\frac{3}{2(1+t)} + \frac{1}{4}\right)t = 1$

$\Rightarrow t^2 + 3t - 4 = 0$ → $t = 1$, → $t = -4$

$\log x = 1$ $\qquad$ $\log x = -4$

$\boxed{x = 10}$ ✓ $\qquad$ $\boxed{x = 0.0001}$ ✗

($\because x > 1$)

Imp

Ques: The coefficient of 3 consecutive terms of $(1+x)^{n+5}$ are in the ratio $5:10:14$, then $n =$ ___.

NOTE: Whenever we have to assume 3 consecutive terms of an Binomial Expansion, we assume them as

T_{r+1}, T_{r+2}, T_{r+3}

Sol^n $T_{r+1} = {}^{n+5}C_r (x)^r$

$T_{r+2} = {}^{n+5}C_{r+1} (x)^{r+1}$

$T_{r+3} = {}^{n+5}C_{r+2} (x)^{r+2}$

Now, $\frac{{}^{n+5}C_{r+1}}{{}^{n+5}C_r} = \frac{10}{5} \Rightarrow \frac{1}{2}$

$\frac{(n+5)-(r+1)+1}{(r+1)} = \frac{1}{2} \Rightarrow$ ① $\boxed{3r = n+3}$

Also, $\frac{{}^{5+n}C_{r+2}}{{}^{5+n}C_{r+1}} = \frac{14}{10} \Rightarrow \frac{7}{5}$

$\frac{(n+5)-(r+2)+1}{(r+2)} = \frac{7}{5} \Rightarrow$ ② $\boxed{12r = 5n+6}$

② − 4 ① ⇒

$(12r - 12r) = (5n - 4n) + (6 - 12)$

$\Rightarrow \boxed{n = 6}$ Ans

Type 3 : Problems Involving Coff. of x^r ⇒

Coff. of x^r in expansion of $(a+x)^n = {}^nC_r \cdot (a)^{n-r}$

Concept: If $\boxed{{}^nC_x = {}^nC_y}$ then, there can be two possible relations b/w x & y

$\boxed{x = y}$ $\qquad$ $\boxed{x + y = n}$

Ques: Smallest Natural No. n, such that the coeff. of x in expansion of $\left(x^2 + \frac{1}{x^3}\right)^n$ is ${}^nC_{23}$, is ____.

Sol^n $T_{r+1} = {}^nC_r (x^2)^{n-r} \left(\frac{1}{x^3}\right)^r$

$= {}^nC_r (x^{2n-5r})$

We want coeff. of x^1 i.e $\boxed{2n - 5r = 1}$

And, $\quad {}^nC_r = {}^nC_{23}$

$r = 23$ $\qquad$ $r + 23 = n$

$2n - 5(23) = 1$ $\qquad$ $2n - 5(n-23) = 1$

$\Rightarrow n = 58$ $\qquad$ $\Rightarrow \boxed{n = 39}$ Ans

Ques The positive value of λ for which the coeff. of x^2 in $x^2\left(\sqrt{x} + \frac{\lambda}{x^2}\right)^{10}$ is 720, is ____.

Sol^n For only x^2

$x^2 \left(x^{1/2} + \lambda \cdot x^{-2}\right)^{10}$

This entire expression should have x^0.

So, $T_{r+1} = {}^{10}C_r (x^{1/2})^{10-r} (\lambda \cdot x^{-2})^r$

$= {}^{10}C_r \left(x^{\frac{10-r}{2} - 2r}\right) \lambda^r$

For x^0, $\frac{10-r}{2} - 2r = 0 \Rightarrow \boxed{r = 2}$

i.e $T_3 = {}^{10}C_2 (x^0) \lambda^2 = 720$

$\frac{10 \times 9}{1 \times 2} \times \lambda^2 = 720$

$\Rightarrow \boxed{\lambda = 4}$ Ans

Type 4 : Term Independent of X / Constant Term ⇒

Ques The term independent of x in expansion of $\left(\frac{x+1}{x^{2/3} - x^{1/3} + 1} - \frac{x-1}{x - \sqrt{x}}\right)^{10}$ is ____.

Sol^n $x + 1 = (x^{1/3})^3 + (1)^3$

$= (x^{1/3} + 1)(x^{2/3} + 1 - x^{1/3})$

So, Expression $= \left(x^{1/3} + 1 - \frac{x-1}{x - x^{1/2}}\right)^{10}$

$= \left(x^{1/3} + 1 - \frac{x-1}{\sqrt{x}(\sqrt{x}-1)}\right)^{10}$

$= \left(x^{1/3} + 1 - \frac{(\sqrt{x}-1)(\sqrt{x}+1)}{\sqrt{x}(\sqrt{x}-1)}\right)^{10}$

$= \left(x^{1/3} - x^{-1/2}\right)^{10}$

So, $T_{r+1} = {}^{10}C_r (x^{1/3})^{10-r} \cdot (-x^{-1/2})^r$

$= {}^{10}C_r \cdot x^{\left(\frac{10-r}{3} - \frac{r}{2}\right)} \cdot (-1)^r$

For independent of x term

$\frac{10-r}{3} - \frac{r}{2} = 0 \Rightarrow 10 - r = \frac{3r}{2}$

$\Rightarrow \boxed{r = 4}$

So, $T_{r+1} = T_5 = {}^{10}C_4 (x)^0 \cdot (-1)^4$

$= \frac{10 \times 9 \times 8 \times 7}{1 \times 2 \times 3 \times 4} \Rightarrow \boxed{210}$ Ans

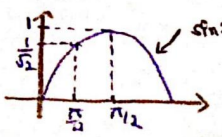
Concept :→

Find Least value of $\frac{{}^{10}C_8 (2)^8}{(\sin 2\theta)^8}$ when

(i) $\theta \in [\pi/16, \pi/4]$

Sol^n $\Rightarrow 2\theta \in \left[\frac{\pi}{4}, \frac{\pi}{2}\right]$

expression will be least when denominator is max.



So, $(\sin 2\theta)_{max} = 1$ when $2\theta \in [\pi/4, \pi/2]$

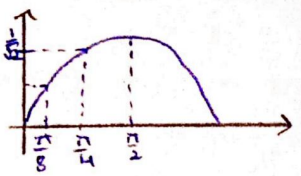
So, $\boxed{\text{Least value} = {}^{10}C_8 (2^8)}$ Ans

(ii) $\theta \in [\pi/16, \pi/8]$

Sol^n $2\theta \in [\pi/8, \pi/4]$

$(\sin 2\theta)_{max} = \frac{1}{\sqrt{2}}$

when $2\theta \in [\pi/8, \pi/4]$



So, $\boxed{\text{Least value} = {}^{10}C_8 \cdot 2^{12}}$ Ans

Ques In the expansion of $\left(\frac{x}{\cos\theta} + \frac{1}{x \sin\theta}\right)^{16}$

$l_1 \rightarrow$ Least value of const. term, $\theta \in \left[\frac{\pi}{8}, \frac{\pi}{4}\right]$

$l_2 \rightarrow$ Least value of const. term, $\theta \in \left[\frac{\pi}{16}, \frac{\pi}{8}\right]$

Then find $l_1 : l_2$.

Sol^n $T_{r+1} = {}^{16}C_r \left(\frac{x}{\cos\theta}\right)^{16-r} \left(\frac{1}{x \sin\theta}\right)^r$

$\left[T_{r+1} = {}^{16}C_r \frac{x^{16-2r}}{(\cos\theta)^{16-r} (\sin\theta)^r}\right]$

For const. term, $16 - 2r = 0 \Rightarrow \boxed{r = 8}$

So, $T_9 = {}^{16}C_8 \frac{1}{(\cos\theta)^8 (\sin\theta)^8} \Rightarrow \frac{{}^{16}C_8 (2)^8}{(\sin 2\theta)^8}$

Now,

$\theta \in \left[\frac{\pi}{8}, \frac{\pi}{4}\right] \Rightarrow 2\theta \in \left[\frac{\pi}{4}, \frac{\pi}{2}\right]$

i.e $(\sin 2\theta)_{max} = 1$

$\boxed{l_1 = {}^{16}C_8 (2)^8}$

$\theta \in [\pi/16, \pi/8] \Rightarrow 2\theta \in \left[\pi/8, \frac{\pi}{4}\right]$

i.e $(\sin 2\theta)_{max} = 1/\sqrt{2}$

$\boxed{l_2 = {}^{16}C_8 (2)^{12}}$

$\frac{l_1}{l_2} = \frac{{}^{16}C_8 (2)^8}{{}^{16}C_8 (2)^{12}} \Rightarrow \frac{2^0}{2^4} \Rightarrow \frac{1}{16}$ Ans

Type 5 : Lengthy Standard Calculations ⇒

→ Always simplify first if possible

→ Higher power wale ko humesha expand karo

→ Repeated expression ko kuch aisa likho General Term,

Ques Find coeff of x^7 in $(1 + 2x - 3x^2)\left(x^2 + \frac{5}{x}\right)^{11}$

Sol^n $= \left(x^2 + \frac{5}{x}\right)^{11} + 2x\left(x^2 + \frac{5}{x}\right)^{11} - 3x^2\left(x^2 + \frac{5}{x}\right)^{11}$

Repeated expression ⇒ $T_{r+1} = {}^{11}C_r (x^2)^{11-r} \left(\frac{5}{x}\right)^r$

$T_{r+1} = {}^{11}C_r (x)^{22-3r} (5)^r$

Coeff of x^7 ⇒ Now, $22 - 3r = 7$ $\qquad$ $22 - 3r = 6$ $\qquad$ $22 - 3r = 5$

$\boxed{r = 5}$ $\qquad$ $r = \frac{16}{3}$ (X) $\qquad$ $r = \frac{17}{3}$ (X)

So, Coeff. of x^7 = $\binom{11}{5}(5)^5$ Ans

Ques Find coeff. of x^5 in $(1-2x-x^2+2x^3)^7$

Sol^n $= (1-2x-x^2+2x^3)^7$

$= [1-2x-x^2(1-2x)]^7$

$= (1-2x)^7 (1-x^2)^7$

Higher power expression ⇒ Expand karenge jaha tak required hoga wahi tak.

$[{}^7C_0(-x)^0 + {}^7C_1(-x^2)^1 + {}^7C_2(-x^2)^2 - - - -]$

$(1-2x)^7$ ⇒ is the repeated expression

$T_{r+1} = {}^7C_r(-2x)^r$

Coeff. of x^5 = $(x^1 \cdot x^4) + (x^3 \cdot x^2) + (x^5 \cdot x^0)$

$= [{}^7C_1(-2)^1 \cdot {}^7C_2] + [{}^7C_3(-2)^3 \cdot {}^7C_1(-1)^1] + [{}^7C_5(-2)^5]$ Ans

Type 6 : Involving Special GP ⇒

→ Power of one term ↓ing & other term ↑ing.

$[a^{(n-1)}.b^0 + a^{(n-2)}.b^1 + a^{(n-3)}.b^2 - - + a^0.b^{(n-1)}]$

First Term ⇒ (a^{n-1}) (a) No. of terms ⇒ n

Common Ratio ⇒ $(\frac{b}{a})$ (r) Sum ⇒ $\frac{a(1-r^n)}{(1-r)}$

Ques Find coefficient of x^5 in

$Y = (x+2)^{n-1} + (x+2)^{n-2}(x+1) - - - (x+1)^{n-1}$

Sol^n $Y = \frac{(x+2)^{n-1}\left[1-\left(\frac{x+1}{x+2}\right)^n\right]}{1-\left(\frac{x+1}{x+2}\right)}$

$\Rightarrow Y = (2+x)^n - (1+x)^n$

$(T_{r+1} = {}^nC_r(2)^{n-r}.x^r)$ $(T_{r+1} = {}^nC_r.x^r)$

Coeff. of $x^r = {}^nC_r(2)^{n-r} - {}^nC_r$ Ans

Type 7 : Hockey Stick Series Appl^n Questions ⇒

Steps to Solve Questions ⇒

1) Find unity in ${}^nC_0$ or ${}^nC_n$

2) Bring unity always at first place; reverse the series if needed.

3) Apply $({}^nC_r + {}^nC_{r-1} = {}^{n+1}C_r)$

Special Series ⇒

1) $({}^{n-1}C_r + {}^{n-2}C_r + {}^{n-3}C_r - - - + {}^rC_r) = {}^nC_{r+1}$

Binomial Theorem-2

2) $({}^nC_0 + {}^{n+1}C_1 + {}^{n+2}C_2 - - - + {}^{n+r}C_r) = {}^{n+r+1}C_r$

Ques If $n \geq 2$ then find sum of

${}^{n+1}C_2 + 2({}^2C_2 + {}^3C_2 + {}^4C_2 + - - - + {}^nC_2)$.

Sol^n ${}^2C_2 + {}^3C_2 + {}^4C_2 + - - - + {}^nC_2$

$\Rightarrow {}^3C_3 + {}^3C_2 + {}^4C_2 - - - + {}^nC_2$ ⇒ ${}^4C_3$ ⇒ ${}^5C_3$ ⇒ ${}^{n+1}C_3$

(We can write ${}^2C_2$ as ${}^3C_3$ as both has value 1)

So, Sum $= {}^{n+1}C_2 + 2({}^{n+1}C_3)$

$= {}^{n+1}C_2 + {}^{n+1}C_3 + {}^{n+1}C_3$

$= {}^{n+2}C_3 + {}^{n+1}C_3$

$= \frac{(n+2)!}{(n-1)!\,(3)!} + \frac{(n+1)!}{(n-2)!\,(3)!}$

$\Rightarrow \frac{(n+2)(n+1)(n)(n-1)!}{(n-1)!\,(3)!} + \frac{(n+1)(n)(n-1)(n-2)!}{(n-2)!\,(3)!}$

$\Rightarrow \frac{(n)(n+1)}{6}\left[\frac{n+2}{1} + \frac{(n-1)}{1}\right] \Rightarrow \frac{n(n+1)(2n+1)}{6}$ Ans

Ques Let m be the smallest +ve integer such that coeff. of x^2 in $(1+x)^2 + (1+x)^3 + - - - - (1+x)^{49} + (1+mx)^{50}$ is $(3n+1)^{51}C_3$ for some +ve integer n, then find n.

Sol^n Coeff. of x^2

$= ({}^2C_2 + {}^3C_2 + {}^4C_2 - - - - {}^{49}C_2) + {}^{50}C_2(m)^2$

$= {}^{50}C_3 + {}^{50}C_2 \cdot m^2 = (3n+1)^{51}C_3$

${}^{50}C_3 + {}^{50}C_2 \cdot m^2 = (3n+1)\left(\frac{51}{3}\right){}^{50}C_2$

$\Rightarrow \frac{{}^{50}C_3}{{}^{50}C_2} + m^2 = 17(3n+1)$

$\frac{50-3+1}{3} + m^2 = 51n + 17$ $\left[\frac{{}^nC_r}{{}^nC_{r-1}} = \frac{n-r+1}{r}\right]$

$\Rightarrow \boxed{m^2 = 51n + 16}$

m → smallest integer

m^2 → Perfect sq., as $(\text{Integer})^2$ = Perfect square

$n=1 \Rightarrow m^2 = 52$ ✗ $n=4 \Rightarrow m^2 = 205$ ✗

$n=2 \Rightarrow m^2 = 103$ ✗ $n=5 \Rightarrow m^2 = 256$ ✓

$n=3 \Rightarrow m^2 = 154$ ✗

$\Rightarrow \boxed{n=5}$ Ans

Type 8 : Kth term from end ⇒

→ Kth term from end is same as $(n-k+2)^{th}$ term from starting.

→ Kth term from end of $(a+x)^n$ is same as Kth term from start of $(x+a)^n$.

Ques Ratio of 7th term from beginning and end in expansion of $(2^{1/3} + 3^{-1/3})^n$ is 1/6, then find value of n.

Sol^n $(2^{1/3} + 3^{-1/3})^n \Rightarrow T_7 = {}^nC_6(2^{1/3})^{n-6}.(3^{-1/3})^6$

$(3^{-1/3} + 2^{1/3})^n \Rightarrow T_7' = {}^nC_7(3^{-1/3})^{n-6}(2^{1/3})^6$

$\frac{T_7}{T_7'} = \frac{(2^{1/3})^{n-6}.(3^{-1/3})^6}{(2^{1/3})^6.(3^{-1/3})^{n-6}} = \frac{1}{6}$

$\Rightarrow (2)^{\frac{n-6}{3}-2}.(3)^{-2-\frac{6-n}{3}} = (2)^{-1}(3)^{-1}$

On Comparing

$\frac{n-6}{3} - 2 = -1$ ⇒ $\boxed{n=9}$ Ans

$\Rightarrow n-6 = 3$

Type 09 : Number of Rational / Irrational Terms

Important Info.

1) (Prime No.)^{Integer} = Rational No.

2) (Prime No.)^{Decimal} = Irrational No.

Approach for Ques

→ Write the General term (T_{r+1})

→ Factorise to prime numbers.

Middle Term

In the expansion of $(a+x)^n$

n = odd ⇒ Two middle terms $T_{\frac{n+1}{2}}$ and $T_{\frac{n+3}{2}}$

n = even ⇒ One middle term $T_{\frac{n+2}{2}}$

Ques In the expansion of $\left(4^{1/3} + \frac{1}{6^{1/4}}\right)^{20}$, find no. of rational & irrational terms and also the middle term.

Sol^n $T_{r+1} = {}^{20}C_r(4^{1/3})^{20-r}.(6^{-1/4})^r$

Imp $= {}^{20}C_r(2)^{\frac{160-11r}{12}}.(3)^{-r/4}$

Always Rational

For these to be Rational, their powers must be integer.

$\left(\frac{-r}{4}\right)$ is integer ⇒ (0, 4, 8, 12, 16, 20)

$\left(\frac{160-11r}{12}\right)$ is integer ⇒ (8, 20)

So, T_9 & T_{21} → Only Rational Terms

Total = 2 ⇒ 21 − 2 ⇒ 19 → Irrational Terms

Middle Term = $T_{\frac{20+2}{2}} \Rightarrow (T_{11})$ ← Irrational Term

NOTE : Imp Formula

$(2n)! = 2^n.(n!)[1.3.5 - - - - .(2n-1)]$

Eg : Different forms of middle term of $\left(x+\frac{1}{x}\right)^{2n}$.

Sol^n Middle Term ⇒ $T_{\frac{2n+2}{2}} \Rightarrow T_{n+1}$

$T_{n+1} = {}^{2n}C_n.(x)^{2n-n}.\left(\frac{1}{x}\right)^n \Rightarrow [T_{n+1} = {}^{2n}C_n]$

So, $\boxed{T_{n+1} = \frac{(2n)!}{n!\,n!}}$ — 1st Form

Now using formula of 2!

$$T_{n+1} = \frac{2^n (1\cdot3\cdot5\cdots(2n-1))}{n!} \rightarrow 2^{nd} \text{ form}$$

$$T_{n+1} = \frac{2\cdot6\cdot10\cdot12\cdots(4n-2)}{n!} \rightarrow 3^{rd} \text{ form}$$

Also,

$$\frac{2n!}{n!\,n!} = \frac{(1\cdot2\cdot3\cdot4\cdots n)(n+1)(n+2)\cdots(2n+n)}{(n!)\quad(1\cdot2\cdot3\cdot4\cdots n)}$$

$$\Rightarrow T_{n+1} = \frac{n+1}{1}\cdot\frac{n+2}{2}\cdot\frac{n+3}{3}\cdots\frac{2n}{n} \rightarrow 4^{th} \text{ form}$$

Type 10: Using AM-GM Inequality ⇒

Que Let $\alpha>0$, $\beta>0$ be such that $\alpha^3+\beta^2=4$. If max. value of const. term of $(\alpha x^{1/9}+\beta x^{-1/6})^{10}$ is $10K$, then $K=$ ___?

Sol^n $T_{r+1} = {}^{10}C_r (\alpha x^{1/9})^{10-r}\cdot(\beta x^{-1/6})^r$

$= {}^{10}C_r (x)^{\frac{10-r}{9}-\frac{r}{6}}\cdot(\alpha)^{10-r}\cdot(\beta)^r$

Constant Term: $\frac{10-r}{9_3} = \frac{r}{6_2} \Rightarrow 20-2r=3r \Rightarrow \boxed{r=4}$

So, $\boxed{T_5 = {}^{10}C_4 (\alpha)^6(\beta)^4}$

↳ It will be max. when $(\alpha^3\cdot\beta^2)^2$ would be maximum

$\because$ AM ≥ GM ⇒ $\frac{\alpha^3+\beta^2}{2} \geq \sqrt{\alpha^3\cdot\beta^2}$

$4 \geq (\alpha^3\cdot\beta^2) \Leftarrow 2 \geq \sqrt{\alpha^3\cdot\beta^2}$

↳ $\boxed{(\alpha^3\cdot\beta^2)_{max}=4}$

So, $(T_5)_{max} = \frac{10\times\overset{3}{\cancel{9}}\times\overset{2}{\cancel{8}}\times7}{1\times2\times3\times4}\times4\times4 = 10K$

$\Rightarrow \boxed{K=336}$ Ans

#Finding Sum of All Coefficients (SOAC):

→ Simply put (all variables = 1) to get SOAC.

Eg: $(2+5x^2-7x^3)^{2000} = (2+5-7)^{2000} = 0^{2000} = 0$ Ans

Eg: If SOAC in $\left(x+\frac{1}{x}\right)^n$ is 64, find term independent of x.

Sol^n $\left(1+\frac{1}{1}\right)^n = 64 \Rightarrow 2^n = (2)^6 \Rightarrow \boxed{n=6}$

$T_{r+1} = {}^6C_r (x)^{6-r}\left(\frac{1}{x}\right)^r \Rightarrow {}^6C_r (x)^{6-2r}$

for constant term

$6-2r=0 \Rightarrow \boxed{r=3} \Rightarrow \boxed{T_4 = {}^6C_3}$ Ans

#Cancellation of Terms:

1) If $(a+x)^n$ & $(a-x)^n$ are added, then only even terms present

$$(a+x)^n + (a-x)^n = 2\left({}^nC_0 a^n\cdot x^0 + {}^nC_2\cdot a^{n-2}\cdot x^2 + {}^nC_4\cdot a^{n-4}\cdot x^4 + \cdots\right)$$

2) If $(a+x)^n$ & $(a-x)^n$ are subtracted, then only odd terms present.

$$(a+x)^n - (a-x)^n = 2\left({}^nC_1 a^{n-1}\cdot x + {}^nC_3\cdot a^{n-3}\cdot x^3 + {}^nC_5\cdot a^{n-5}\cdot x^5 + \cdots\right)$$

Que α & β are coeff of x^4 & x^2 in expansion of $(x+\sqrt{x^2-1})^6 + (x-\sqrt{x^2-1})^6$, then find relation b/w α & β.

Sol^n $(x+\sqrt{x^2-1})^6 + (x-\sqrt{x^2-1})^6$

$= 2\left[{}^6C_0 x^6 + {}^6C_2 x^4 (x^2-1)^2 + {}^6C_4\cdot x^2(x^2-1)^2 + {}^6C_6\cdot x^0 (x^2-1)^3\right]$

$= 2\left[32x^6 - 48x^4 + 18x^2 - 1\right]$

$\boxed{\alpha = -96}$ $\boxed{\beta = 36}$

So, $\boxed{\alpha-\beta=-132}$ Ans

#Divisibility Problems:

Suppose we are given,

$(1+\alpha)^n = \left(1 + {}^nC_1\alpha + {}^nC_2\alpha^2 + \cdots + {}^nC_n\alpha^n\right)$

This expression is not divisible by α because of (1) and Remainder = 1.

$\Rightarrow (1+\alpha)^n - 1 = \left({}^nC_1\alpha + {}^nC_2\alpha^2 + \cdots + {}^nC_n\alpha^n\right)$

This expression is divisible by 'α' as every term has α in it

Jisse divide karenge, wahi form/no. banane ke koshish karenge.

Eg: Is (8^n-1-7n) divisible by 49?

Sol^n $= (1+7)^n - 1 - 7n$

$= \left(1 + {}^nC_1(7) + {}^nC_2(7)^2 + \cdots + {}^nC_n(7)^n\right) - 1 - 7n$

$= \left[{}^nC_2(7)^2 + {}^nC_3(7)^3 + \cdots + {}^nC_n(7)^n\right]$

↳ this expression is divisible by $(7)^2$. Yes.

#Use of Functions:

Greatest Integer Function ⇒

$[\text{Integer}] = \text{Integer}$; $[\text{Non-Integer}] =$ Just left side integer on no. line

Eg: $[-2.1] = -3$ $\quad [\pi] = 3$

$[2.1] = 2$ $\quad [-1.2] = -2$

Fractional part Function ⇒

$\{x\} = x$ when $x \in [0,1)$

$\{x + \text{Integer}\} = \{x\}$

And, Imp $\boxed{\{x\} = x - [x]}$

$\{x\} \in [0,1)$ always

Eg: If $\{x\}$ denotes fractional part of x, then $\left\{\frac{3^{200}}{8}\right\}$ is equal to?

Sol^n Since there is 8 in division, we try to make 8 in numerator.

$$\left\{\frac{3^{200}}{8}\right\} = \left\{\frac{(3^2)^{100}}{8}\right\} = \left\{\frac{(8+1)^{100}}{8}\right\}$$

$$= \left\{\frac{1 + {}^{100}C_1(8) + {}^{100}C_2(8)^2 + \cdots + {}^{100}C_{100}(8)^{100}}{8}\right\}$$

$$= \left\{\frac{1}{8} + \left({}^{100}C_1(8) \cdots + {}^{100}C_{100}(8)^{100}\right)\right\}$$

↳ Integer.

$\therefore \{x + \text{integer}\} = \{x\}$

$\Rightarrow \left\{\frac{3^{200}}{8}\right\} = \left\{\frac{1}{8}\right\} = \frac{1}{8}$ Ans

Remainder Ka Gyaan: → divided

When a number is divisible by p and leaves a remainder q, then the no. can be written as –

$\boxed{\text{Number} = (pK+q)}$ where $(K \in \text{Integer})$

Eg: $26 = 4(6)+2$

$73 = 8(9)+1$

Que If the remainder, when x is divided by 4 is 3, then the remainder when $(2020+x)^{2022}$ is divided by 8 is ___.

Sol^n $[x = 4K+3]$

↳ $(2020+4K+3)^{2022}$

$= (4M+3)^{2022} \Rightarrow (4M+4-1)^{2022}$

$= (4\lambda-1)^{2022}$ $\quad [4M+4 = 4\lambda \in \text{Integer}]$

$2020+4K = 4(505+K) = 4M$; $M \rightarrow$ Integer

Now,

$(4\lambda-1)^{2022} = {}^{2022}C_0(4\lambda)^{2022} - {}^{2022}C_1(4\lambda)^{2021} + {}^{2022}C_2(4\lambda)^{2020} - \cdots - {}^{2022}C_{2021}(4\lambda)^1 + {}^{2022}C_{2022}$

All terms of expansion have 4 except last this. Hence all terms would be divisible by 8, except 1st.

Remainder $= {}^{2022}C_{2022} = 1$ Ans

#Greatest Binomial Coefficient:

Out of ${}^nC_0, {}^nC_1, {}^nC_2 \cdots {}^nC_{n-1}, {}^nC_n$ the greatest Binomial coefficient is

$\boxed{n = \text{even} \Rightarrow {}^nC_{n/2}}$

$\boxed{n = \text{odd} \Rightarrow {}^nC_{\frac{(n-1)}{2}} \text{ and } {}^nC_{\frac{(n+1)}{2}}}$

Binomial Theorem-3

Ques If a,b,c are greatest values of $^{19}C_p$, $^{20}C_q$ and $^{21}C_r$, then find their ratios.

Soln $a = {}^{19}C_9 = {}^{19}C_{10} \Rightarrow a/{}^{19}C_9 = 1$

$b = {}^{20}C_{10} \Rightarrow b/{}^{20}C_{10} = 1$

$c = {}^{21}C_{10} = {}^{21}C_{11} \Rightarrow c/{}^{21}C_{11} = 1$

So, $\frac{a}{{}^{19}C_9} = \frac{b}{{}^{20}C_{10}} = \frac{c}{{}^{21}C_{11}}$

$\Rightarrow \frac{a}{{}^{19}C_9} = \frac{b}{\left(\frac{20}{10}\right){}^{19}C_9} = \frac{c}{\left(\frac{21}{11}\right)\left(\frac{20}{10}\right){}^{19}C_9}$

$\Rightarrow \boxed{\frac{a}{11} = \frac{b}{22} = \frac{c}{42}}$ Ans

Questions on series :

1) Basic Substitution :→

$(1+x)^n = {}^nC_0 + {}^nC_1 x + {}^nC_2 x^2 \cdots + {}^nC_n x^n$

(x=1) $\Rightarrow \boxed{2^n = {}^nC_0 + {}^nC_1 + {}^nC_2 + {}^nC_3 \cdots + {}^nC_n}$

(x=-1) $\Rightarrow \boxed{0 = {}^nC_0 - {}^nC_1 + {}^nC_2 - {}^nC_3 \cdots + {}^nC_n(-1)^n}$

↳ When alternate plus minus.

Also, $\boxed{2^{n-1} = {}^nC_0 + {}^nC_2 + {}^nC_4 \cdots = {}^nC_1 + {}^nC_3 + {}^nC_5 \cdots}$

↳ Sum of even or odd Binomial coeff.

Ques If $\sum_{r=0}^{25} ({}^{50}C_r \cdot {}^{50-r}C_{25-r}) = K({}^{50}C_{25})$, then K is equal to ____.

Soln ${}^{50}C_r \cdot {}^{50-r}C_{25-r} = \frac{50!}{r!(50-r)!} \times \frac{(50-r)!}{(25-r)!(25)!}$

$= \frac{50!}{25!} \cdot \frac{1}{(25-r)!r!} \Rightarrow \left(\frac{50!}{25!\,25!}\right)\frac{25!}{(25-r)!r!}$

$= {}^{50}C_{25} \cdot {}^{25}C_r$

Now, $\sum_{r=0}^{25} {}^{50}C_{25} \cdot {}^{25}C_r = K({}^{50}C_{25})$

${}^{50}C_{25} \cdot (2^{25}) = K({}^{50}C_{25}) \Rightarrow \boxed{K = 2^{25}}$ Ans

2) Using Differentiation :→

$(1+x)^n = {}^nC_0 + {}^nC_1 x + {}^nC_2 x^2 + {}^nC_3 x^3 \cdots$

$\frac{d}{dx}$ $n(1+x)^{n-1} = 0 \cdot {}^nC_0 + 1 \cdot {}^nC_1 + 2 \cdot {}^nC_2 x + \cdots$

(x=1) $\Rightarrow \boxed{n(2)^{n-1} = 0 \cdot {}^nC_0 + 1 \cdot {}^nC_1 + 2 \cdot {}^nC_2 + 3 \cdot {}^nC_3 \cdots}$

↳ When coeff. multiplied with r.

(x=-1) $\Rightarrow \boxed{0 = 0 \cdot {}^nC_0 - 1 \cdot {}^nC_1 + 2 \cdot {}^nC_2 - 3 \cdot {}^nC_3 + \cdots}$

↳ When coeff multiplied with r & alternate plus minus

Ques Value of $-{}^{15}C_1 + {}^{15}C_2 - {}^{15}C_3 \cdots - 15 \cdot {}^{15}C_{15} + {}^{14}C_1 + {}^{14}C_3 + {}^{14}C_5 \cdots + {}^{14}C_{11}$ is ____.

Soln $-{}^{15}C_1 + 2{}^{15}C_2 - \frac{1}{3}{}^{15}C_3 \cdots - 15 \cdot {}^{15}C_{15} = 0$

Now,

$({}^{14}C_1 + {}^{14}C_3 + {}^{14}C_5 + {}^{14}C_7 \cdots + {}^{14}C_{11} + {}^{14}C_{13}) - {}^{14}C_{13}$

$= 2^{14-1} - {}^{14}C_{13} \Rightarrow \boxed{2^{13} - 14}$ Ans

Ques Let $m,n \in N$ and $\gcd(2,n) = 1$. If $30\binom{30}{0} + 29\binom{30}{1} + \cdots 2\binom{30}{28} + 1\binom{30}{29} = n \cdot 2^m$ then n+m is equal to ____.

Soln $= 30 \cdot {}^{30}C_0 + 29 \cdot {}^{30}C_1 + \cdots + 1 \cdot {}^{30}C_{29}$

(${}^{30}C_{30}$, ${}^{30}C_{29}$, ${}^{30}C_1$)

$= 1 \cdot {}^{30}C_1 + 2 \cdot {}^{30}C_2 + \cdots 29 \cdot {}^{30}C_{29} + 30 \cdot {}^{30}C_{30}$

$= 30 \cdot 2^{29} = n \cdot 2^m$

But here $\gcd(30,2) = 2$ and not 1, so we can't take n=30 and m=29

$\Rightarrow \frac{15}{2} \cdot 2^{29} = n \cdot 2^m \Rightarrow 15 \cdot 2^{28} = n \cdot 2^m$

Now, $\gcd(15,2) = 1$ (✓)

Hence, n=15 & m=28 $\Rightarrow \boxed{m+n = 43}$ Ans

Note: Sometimes, we will have to write the general term by observing pattern

Ques If $2^{0}C_1 + (2^2)\,{}^{20}C_2 + (3^2)\,{}^{20}C_3 + \cdots + (20^2)\,{}^{20}C_{20} = A(2^\beta)$ then find A & β.

Soln series $= \sum_{K=0}^{20} (K^2) \cdot {}^{20}C_K$

To lower down the degree of summation variable in product, we always use ${}^nC_r = \frac{n}{r}\,{}^{n-1}C_{r-1}$

$= 20 \sum_{K=1}^{20} (K)\,{}^{19}C_{K-1}$ [Adjusted value of limits acc. to change in ${}^nC_r$]

$= 20\left[1 \cdot {}^{19}C_0 + 2 \cdot {}^{19}C_1 + 3 \cdot {}^{19}C_2 \cdots + 20 \cdot {}^{19}C_{19}\right]$

$= 20\left[2^{18} \cdot 19 + 2^{19}\right] \Rightarrow (420)\,2^{18}$ Ans

Derivation of Formula :→

$(1+x)^n = {}^nC_0 x^0 + {}^nC_1 x + \cdots {}^nC_n x^n$

$x(1+x)^n = {}^nC_0 x + {}^nC_1 x^2 + \cdots {}^nC_n x^{n+1}$

(Diff. w.r.t. x)

$x \cdot n(1+x)^{n-1} + (1+x)^n = 1 \cdot {}^nC_0 + 2 \cdot {}^nC_1 x + 3 \cdot {}^nC_2 x^2 \cdots (n+1)\,{}^nC_n x^n$

(x=1) $\Rightarrow \boxed{(2^{n-1})n + 2^n = 1 \cdot {}^nC_0 + 2 \cdot {}^nC_1 \cdots + (n+1)\,{}^nC_n}$

↳ When series is of form $(r+1) \cdot {}^nC_r$

3) Using Integration :→

$(1+x)^n = {}^nC_0 x^0 + {}^nC_1 x + {}^nC_2 x^2 + \cdots {}^nC_n x^n$

$\int (1+x)^n dx = \int ({}^nC_0 + {}^nC_1 x \cdots + {}^nC_n x^n) dx$

$\frac{(1+x)^{n+1}}{n+1} = {}^nC_0 x + {}^nC_1 \frac{x^2}{2} + {}^nC_2 \frac{x^3}{3} \cdots$

Put x=0 ↳ $\boxed{C = \frac{1}{n+1}}$ to find const. of integration

$\Rightarrow \frac{(1+x)^{n+1}}{n+1} = {}^nC_0 x + \frac{{}^nC_1 x^2}{2} + \cdots \frac{{}^nC_n x^{n+1}}{n+1} + \left(\frac{1}{n+1}\right)$

(Put x=1) $\boxed{\frac{2^{n+1}}{n+1} - \frac{1}{n+1} = \frac{{}^nC_0}{1} + \frac{{}^nC_1}{2} + \frac{{}^nC_2}{3} \cdots}$

↳ When Binomial coeff. has one more than the Binomial Variable in division

4) Fedric Karl Gauss (FKG) Method :→

(i) Assume given series as S

(ii) S ko ek baar seedha likho, ek baar ulta

(iii) Ulta series pe ${}^nC_r = {}^nC_{n-r}$ lagao.

(iv) Dono seedha aur ulti S ko add karado

(v) Solve further acc. to question.

Ques If $a_n = \sum_{r=0}^{n} \frac{1}{{}^nC_r}$ find value of $\sum_{r=0}^{n} \frac{r}{{}^nC_r}$.

Soln $S = \frac{0}{{}^nC_0} + \frac{1}{{}^nC_1} + \frac{2}{{}^nC_2} \cdots + \frac{n-1}{{}^nC_{n-1}} + \frac{n}{{}^nC_n}$

(${}^nC_r = {}^nC_{n-r}$) $S = \frac{n}{{}^nC_n} + \frac{n-1}{{}^nC_{n-1}} + \cdots \frac{2}{{}^nC_2} + \frac{1}{{}^nC_1} + \frac{0}{{}^nC_0}$

(+) $S = \frac{n}{{}^nC_0} + \frac{n-1}{{}^nC_1} + \frac{n-2}{{}^nC_2} \cdots \frac{1}{{}^nC_{n-1}} + \frac{0}{{}^nC_n}$

$2S = \frac{n}{{}^nC_0} + \frac{n}{{}^nC_1} + \frac{n}{{}^nC_2} \cdots \frac{n}{{}^nC_{n-1}} + \frac{n}{{}^nC_n}$

$2S = n \sum_{r=0}^{n} \left(\frac{1}{{}^nC_r}\right) \Rightarrow n \cdot a_n$

$\Rightarrow \boxed{S = \frac{n \cdot a_n}{2}}$ Ans

Ques Find $\sum_{r=0}^{n} \left(\frac{r^2}{r+1}\right) \cdot {}^nC_r$.

Soln $\left(\frac{r^2}{r+1}\right){}^nC_r = \left[\frac{(r^2-1)+1}{r+1}\right] \cdot {}^nC_r$

$\Rightarrow \left(r-1+\frac{1}{r+1}\right){}^nC_r \Rightarrow (r-1)\,{}^nC_r + \frac{{}^nC_r}{r+1}$

$\Rightarrow \underbrace{r \cdot {}^nC_r}_{n(2)^{n-1}} - \underbrace{{}^nC_r}_{2^n} + \underbrace{\frac{{}^nC_r}{r+1}}_{\frac{2^n-1}{(n+1)}}$ Ans

Bino-Binomial Series :

↳ When two Binomial coefficients are present in product.

By FHCG Method

$(1+x)^n = {}^nC_0 + {}^nC_1 x + {}^nC_2 x^2 - - - + {}^nC_{n-1} x^{n-1} + \dots$

$x(1+x)^n = {}^nC_0 x^n + {}^nC_1 x^{n-1} + {}^nC_2 x^{n-2} - - - {}^nC_n x^0$

$(1+x)^{2n} =$ ~~~ + ~~~ + ~~~ + ~~~

→ We will get many terms by multiplying.

→ If we expand L.H.S as ${}^{2n}C_0 + {}^{2n}C_1 x \dots$ and compare with RHS, then we get

1) $({}^nC_0)^2 + ({}^nC_1)^2 + ({}^nC_2)^2 - - - + ({}^nC_n)^2 = {}^{2n}C_n$

↳ when square of Binomial coeff. given

2) ${}^nC_1 \cdot {}^nC_2 + {}^nC_2 \cdot {}^nC_3 - - - + {}^nC_{n-1} \cdot {}^nC_n = {}^{2n}C_{n-1} = {}^{2n}C_{n+1}$

↳ when product of two adjacent B.C. present

Ques Let $X = ({}^{10}C_1)^2 + 2({}^{10}C_2)^2 + 3({}^{10}C_3)^2 + \dots + ({}^{10}C_{10})^2$ then find $\frac{1}{1430} X$.

Sol^n By FHCG Method

$X = 0 \cdot ({}^{10}C_0)^2 + 1({}^{10}C_1)^2 + 2({}^{10}C_2)^2 - - - + 10({}^{10}C_{10})^2$

$+ X = 10({}^{10}C_0)^2 + 9({}^{10}C_1)^2 + 8({}^{10}C_2)^2 - - - + 0({}^{10}C_{10})^2$

$2X = 10\left[({}^{10}C_0)^2 + ({}^{10}C_1)^2 + ({}^{10}C_2)^2 - - - + ({}^{10}C_{10})^2\right]$

$2X = 10 \cdot {}^{20}C_{10} \Rightarrow \boxed{X = 5 \cdot {}^{20}C_{10}}$

So, $\frac{1}{1430} X \Rightarrow \frac{1}{\cancel{1430}\ 286} \times 5 \times {}^{20}C_{10}$

$\boxed{\frac{X}{1430} = 646}$ Ans

④ Bino-Binomial series where sum of lower suffix is constant :→

$\boxed{{}^nC_0 \cdot {}^mC_K + {}^nC_1 \cdot {}^mC_{K-1} + {}^nC_2 \cdot {}^mC_{K-2} - - - - - {}^nC_K \cdot {}^mC_0 = {}^{m+n}C_K}$

↳ Coeff of x^K in $(1+x)^{m+n}$

Ques Let A_r, B_r & C_r denote coeff. of x^r in the expansion of $(1+x)^{10}$, $(1+x)^{20}$ & $(1+x)^{30}$ respectively. Then find $\sum_{r=1}^{10} A_r (B_{10} B_r - C_{10} \cdot A_r)$.

Sol^n $A_r = {}^{10}C_r \quad B_r = {}^{20}C_r \quad C_r = {}^{30}C_r$

$B_{10} = {}^{20}C_{10} \quad C_{10} = {}^{30}C_{10}$

Now,

$A_r (B_{10} B_r - C_{10} \cdot A_r) = {}^{10}C_r \left({}^{20}C_{10} \cdot {}^{20}C_r - {}^{30}C_{10} {}^{10}C_r\right)$

$\Rightarrow \sum_{r=1}^{10} {}^{20}C_{10} \cdot {}^{10}C_r \cdot {}^{20}C_r - \sum_{r=1}^{10} {}^{10}C_r \cdot {}^{10}C_r \cdot {}^{30}C_{10}$

$= {}^{20}C_{10}\left[\sum_{r=1}^{10} {}^{10}C_r \cdot {}^{20}C_r\right] - {}^{30}C_{10}\left[\sum_{r=1}^{10} ({}^{10}C_r)^2\right]$

$\Downarrow$ $\qquad$ $\Downarrow$

${}^{20}C_{10}\left[\sum_{r=0}^{10} {}^{10}C_r \cdot {}^{20}C_{20-r}\right]$ $\qquad$ $({}^{20}C_{10} - 1)$

Due to $\sum ({}^{10}C_0)^2$

$= {}^{20}C_{10}({}^{30}C_{20} - 1) - {}^{30}C_{10}({}^{20}C_{10} - 1)$

Due to $(= {}^{20}C_{20} \cdot {}^{10}C_0)$

$= {}^{20}C_{10}({}^{30}C_{20} - 1) - {}^{30}C_{10} \cdot {}^{20}C_{10} + {}^{30}C_{10}$

$= {}^{30}C_{10} - {}^{20}C_{10} \Rightarrow \boxed{C_{10} - B_{10}}$ Ans

Multinomial Theorem :

For $(a+b+c)^n$, the general term is written as

$$\text{Gen. Term} = \frac{n!}{r_1! r_2! r_3!} (a)^{r_1} (b)^{r_2} (c)^{r_3}$$

where, $(r_1 + r_2 + r_3 = n)$

→ For a given term, if $r_1 + r_2 + r_3 \neq n$ then no such term exist for given $(a+b+c)^n$.

Ques Find coefficient of $a^3 b^4 c^4$ in $(ab + bc + ca)^7$.

Sol^n Gen. Term $= \frac{7!}{r_1! r_2! r_3!} (ab)^{r_1} (bc)^{r_2} (ca)^{r_3}$

$= \frac{7!}{r_1! r_2! r_3!} (a)^{r_1+r_3} (b)^{r_1+r_2} (c)^{r_2+r_3}$

Now, $r_1 + r_2 + r_3 = 7$

$r_1 + r_3 = 3 \Rightarrow \boxed{r_1 = 3}$

$r_2 + r_3 = 4 \Rightarrow \boxed{r_2 = 4}$

$r_1 + r_2 = 7 \Rightarrow \boxed{r_3 = 0}$

So,

Coeff. $= \frac{7!}{3!\,4!\,0!} \Rightarrow \frac{7 \times 6 \times 5}{1 \times 2 \times 3} \Rightarrow 35$ Ans

Ques Coeff of x^7 in $(1 + 3x - 2x^3)^{10}$ is...

Sol^n Gen. Term $= \frac{10!}{r_1! r_2! r_3!} (1)^{r_1} (3x)^{r_2} \left(\frac{-2}{x^3}\right)^{r_3}$

$= \frac{10!}{r_1! r_2! r_3!} (3^{r_2} * (-2)^{r_3}) x^{r_2 + 3r_3}$

Now, $r_1 + r_2 + r_3 = 10$

$r_2 + 3r_3 = 7$

Since r_1, r_2 & r_3 can only be whole numbers lesser than or equal to 10

If $r_3 = 0 \Rightarrow r_2 = 7$ & $r_1 = 3$

$r_3 = 1 \Rightarrow r_2 = 4$ & $r_1 = 5$

$r_3 = 2 \Rightarrow r_2 = 1$ & $r_1 = 7$

Hence,

Coeff. of $x^7 = \frac{10!}{7!\,1!\,2!} (3)^1 (-2)^2$

$+ \frac{10!}{5!\,4!\,1!} (3)^4 (-2)^1 + \frac{10!}{3!\,7!\,0!} (3)^7 (-2)^0$

Ans